BTS – VI – 04.15 –0103 Reg. No. | | | | | | | | |

A

B.Tech. Degree VI Semester Examination April 2015

ME 1601 DYNAMICS OF MACHINERY
(*2012 Scheme*)

Time : 3 Hours Maximum Marks : 100

PART A
(Answer *ALL* questions)

(8 × 5 = 40)

I. (a) Explain how friction in sliding and turning pairs are taken in to account in the force analysis of mechanisms.

(b) Explain what is meant by equivalent dynamic system.

(c) Differentiate between flywheel and governor. What is the significance of turning moment diagram?

(d) Write short notes on different types of brakes.

(e) Explain static balancing and dynamic balancing.

(f) How are in line engines are balanced?

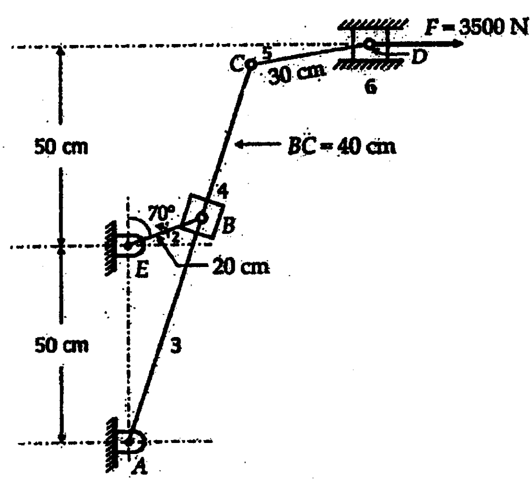
(g) What are the advantages and disadvantages of chain drive over belt drive?

(h) Explain the following terms with reference to a governor.
   (i) Sensitiveness
   (ii) Hunting.

PART B

(4 × 15 = 60)

II. For the static equilibrium of the quick return motion mechanism shown in figure, find the torque required on link 2 (EB) for equilibrium for a force of 3500 N on the slider. Angle EB with the vertical is 70°. Length of link AC is 100 cm.



OR

III. A horizontal steam engine running at 240 rpm has a bore of 20 cm and stroke of 36 cm. The piston rod is 2 cm in diameter and connecting rod length is 90 cm, The mass of the reciprocating parts is 7 kg, and the frictional resistance is equivalent to a force of 500 N . The mean pressure being 50 x 10^2 N/mm^2 on the cover side and 1x 10^2 N/mm^2 on crank side. Determine the following when the crank is at 120° from I.D.C.
   (i) Thrust on the connecting rod.
   (ii) Thrust on the cylinder walls.
   (iii) Load on the crank shaft bearing.
   (iv) Turning moment on the crank shaft.

(P.T.O.)

IV. (a) The torque delivered by a two stroke engine is represented by $T = (1000 + 300 \sin 2\theta - 500 \cos 2\theta)$ N.m where θ is the angle turned by the crank from the inner dead centre. The engine speed is 250 rpm. The mass of the fly wheel is 400 kg, and the radius of gyration is 400 mm. Determine

(i) The power developed.

(ii) The total percentage fluctuation of speed.

(iii) The angular acceleration of flywheel when the crank has rotated through an angle of 60° from inner dead centre.

(iv) The maximum angular acceleration and retardation of the flywheel.

OR

V. The turbine rotor of a ship having a mass of 200 kg, rotates at 2000 rpm, and its radius of gyration is 0.30 m. If the rotation of the rotor is clockwise looking from the aft, determine the gyroscopic couple set by the rotor when

(i) ship takes a left hand turn at a radius of 300 metres at a speed of 30 km/hr.

(ii) ship pitches with the bow rising at an angular velocity of 1 rad/sec.

(iii) ship rolls at an angular velocity of 0.1 rad/sec.

VI. A rotating shaft carries four unbalanced masses 18 kg, 14 kg, 16 kg, and 12 kg, at radii 5 cm, 6 cm, 7 cm and 6 cm respectively. The 2^{nd}, 3^{rd} and 4^{th} masses revolve in planes 8cm, 16 cm, and 28 cm, respectively measured from the plane of the first mass and are angularly located at 60°, 135° and 270° respectively measured anticlockwise from the first mass looking from this mass end of the shaft. The shaft is dynamically balanced by two masses, both located at 5 cm, radii and revolving in planes midway between those of 1^{st} and 2^{nd} masses and midway between those of 3^{rd} and 4^{th} masses. Determine graphically or otherwise, the magnitudes of the masses and their respective angular positions.

OR

VII. A four crank engine has the two outer cranks set at 120° to each other and their reciprocating masses are each 400 kg,. The distance between the planes of rotation of adjacent cranks are 450 mm, 750 mm, and 600 mm,. If the engine is to be in complete primary balance, find the reciprocating mass, and the relative angular position for each of the inner cranks. If the length of each crank is 300 mm, the length of each connecting rod is 1.2 m and the speed of rotation is 240 rpm, what is the maximum secondary unbalanced force?

VIII. An open belt running over two pulleys 240 mm and 600 mm diameter connects two parallel shafts 3 metres apart and transmits 4 kW from the smaller pulley that rotates at 300 rpm. Coefficient of friction between the belt and the pulley is 0.3 and the safe working tension is 10 N per mm width. Determine (i) minimum width of the belt (ii) initial belt tension (iii) length of the belt required.

OR

IX. Explain the working of the following governors.

(i) Watt governor

(ii) Porter governor

(iii) Hartnell governor.

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